

SOC Alert Investigation using LetsDefend SIEM

Project Overview

This project demonstrates the complete investigation workflow of a phishing-related SIEM alert using the LetsDefend platform. The objective was to follow a structured SOC analyst process—from detecting the alert to analyzing evidence and determining whether the incident was a true positive or false positive.

The investigation focused on understanding how a Security Operations Center (SOC) analyst monitors alerts, collects evidence, performs log analysis, investigates suspicious emails, and documents findings before closing the incident.

Learning Objectives

During this project, I focused on developing practical SOC L1 investigation skills by working through a real-world alert scenario.

The main objectives were:

- Become familiar with the LetsDefend SIEM platform and SOC workflow.
 - Learn how security alerts are monitored, assigned, investigated, and closed.
 - Analyze Windows logs and network activity to identify suspicious behavior.
 - Investigate phishing emails using email security tools.
 - Verify whether users interacted with malicious links or downloaded harmful files.
 - Improve incident investigation and documentation skills.
 - Understand how structured playbooks help analysts investigate alerts consistently.
 - Build confidence in identifying true positives and false positives.
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Investigation Scenario

Alert Name: SOC282 – Phishing Alert – Deceptive Mail Detected

Event ID: 257

Alert Category: Phishing

Objective:

Investigate a phishing alert, analyze the related evidence, determine whether malicious activity occurred, and document the findings before closing the incident.

Investigation Workflow

1. Alert Detection

The investigation started from the **Monitoring** dashboard, where all newly generated security alerts are displayed.

The dashboard is divided into three sections:

- Main Channel
- Investigation Channel
- Closed Alerts

 LetsDefend

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 Monitoring

 Log Management

 Case Management

 Endpoint Security

 Email Security

 Threat Intel

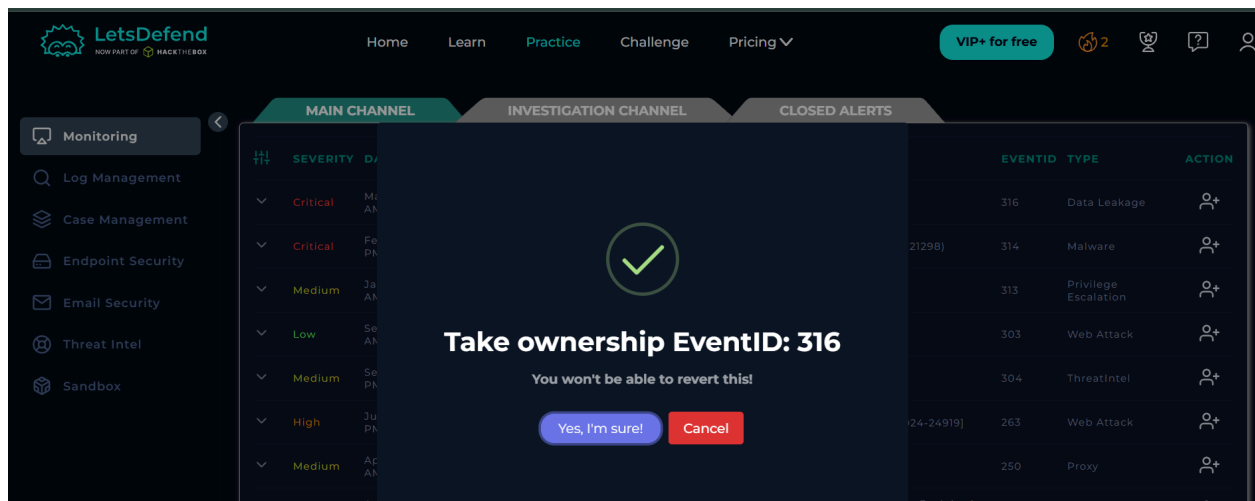
 Sandbox

MAIN CHANNEL

INVESTIGATION CHANNEL

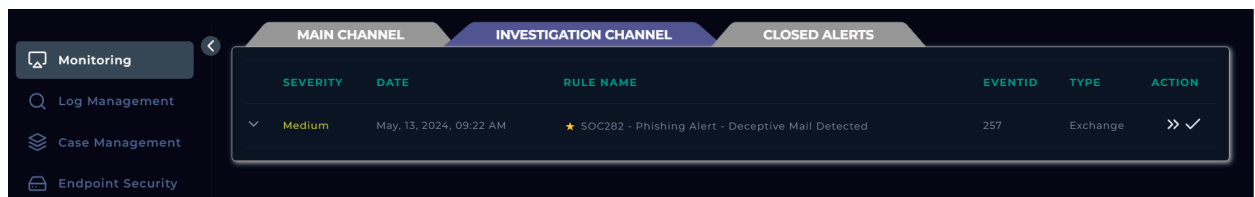
CLOSED ALERTS

TTT	SEVERITY	DATE	RULE NAME	EVENTID	TYPE	ACTION
▼	Critical	Mar, 13, 2025, 09:44 AM	SOC338 - Lumma Stealer - DLL Side-Loading via Click Fix Phishing	316	Data Leakage	
▼	Critical	Feb, 04, 2025, 04:18 PM	SOC336 - Windows OLE Zero-Click RCE Exploitation Detected [CVE-2025-21298]	314	Malware	
▼	Medium	Jan, 22, 2025, 02:37 AM	SOC335 - CVE-2024-49138 Exploitation Detected	313	Privilege Escalation	
▼	Low	Sep, 24, 2024, 08:21 AM	SOC325 - Unauthorized Cloud Region Access Attempt Detected	303	Web Attack	
▼	Medium	Sep, 17, 2024, 12:05 PM	SOC326 - Impersonating Domain MX Record Change Detected	304	ThreatIntel	
▼	High	Jun, 06, 2024, 03:12 PM	★ SOC287 - Arbitrary File Read on Checkpoint Security Gateway [CVE-2024-24919]	263	Web Attack	
▼	Medium	Apr, 19, 2024, 08:23 AM	SOC275 - Application Token Steal Attempt Detected	250	Proxy	
▼	Critical	Apr, 18, 2024, 03:09 AM	★ SOC274 - Palo Alto Networks PAN-OS Command Injection Vulnerability Exploitation [CVE-2024-3400]	249	Web Attack	
▼	Medium	Mar, 14, 2024, 05:23 PM	SOC153 - Suspicious Powershell Script Executed	238	Malware	
▼	High	Mar, 07, 2024, 12:51 PM	SOC127 - SQL Injection Detected	235	Web Attack	

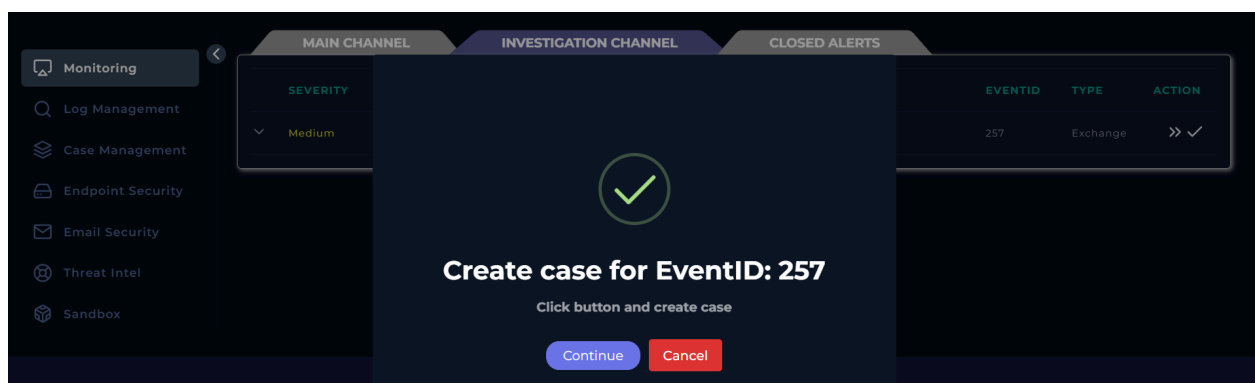


2. Case Creation

Once the alert was assigned, it moved to the **Investigation Channel**.



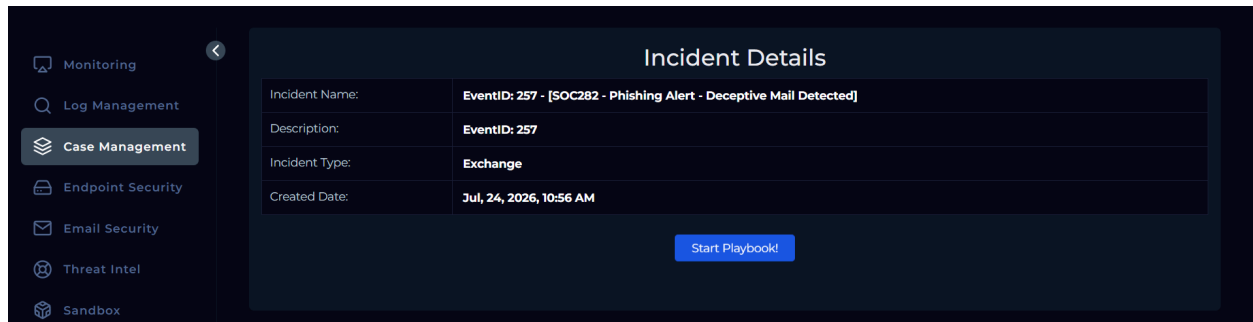
A new case was created, which automatically opened the **Case Management** page.



This page contains all investigation details and assigns a predefined playbook that guides the analyst through every investigation step.

3. Playbook Execution

The playbook provides a structured investigation process to ensure every alert is analyzed consistently. The playbook consists of multiple checkpoints and step-by-step guidance that help analysts investigate alerts accurately and consistently.



Following the playbook helped me:

- Review the alert details.
- Collect relevant evidence.
- Perform email analysis.
- Analyze endpoint information.
- Review logs.
- Validate suspicious activity.
- Record investigation findings.

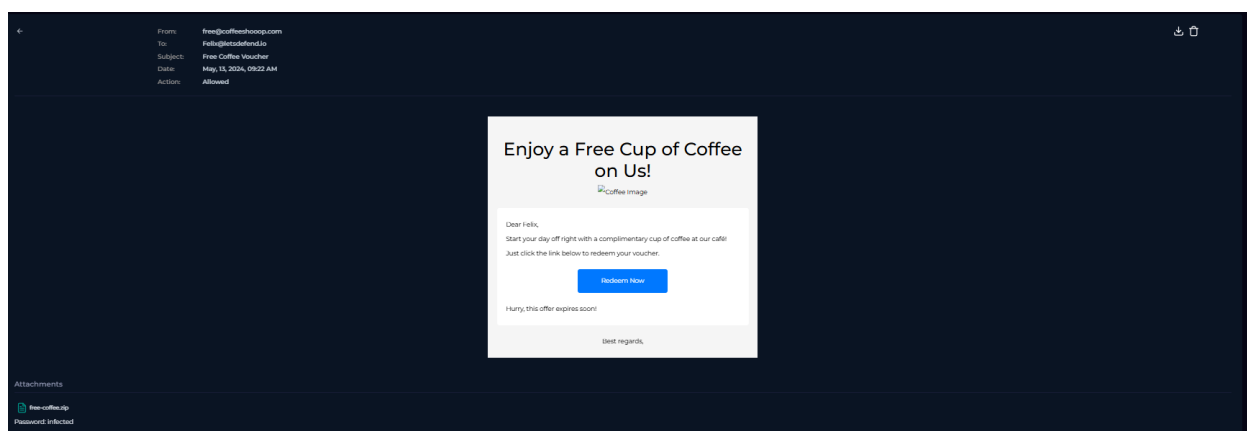
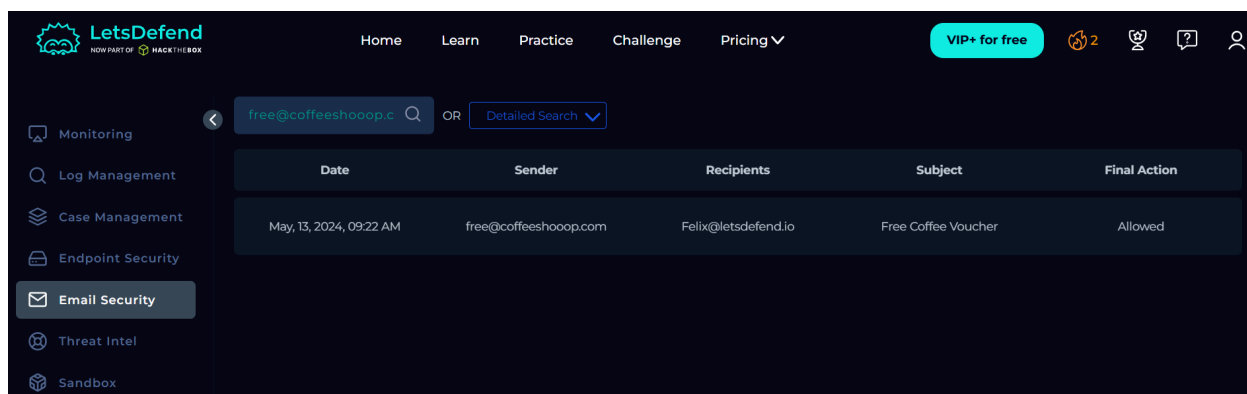
Using a playbook reduces the chances of missing important evidence during an investigation.

4. Email Analysis

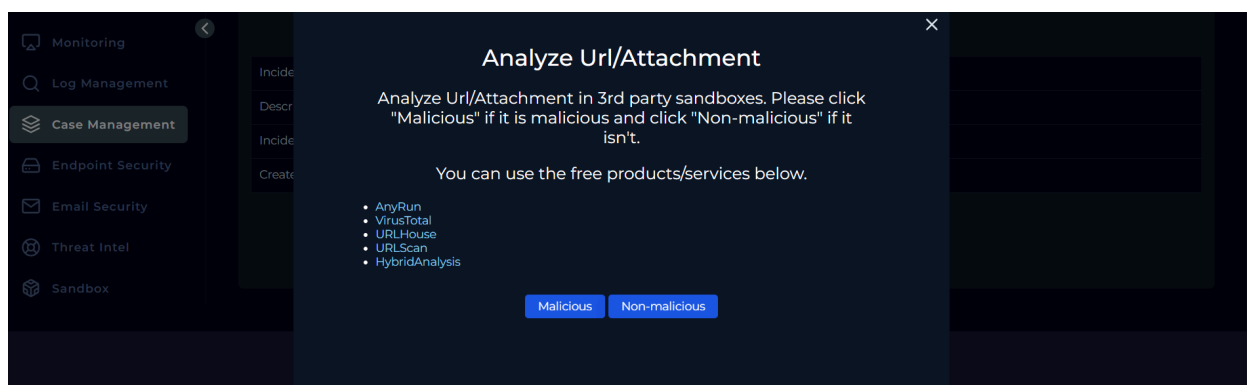
The next step was to examine the suspicious email.

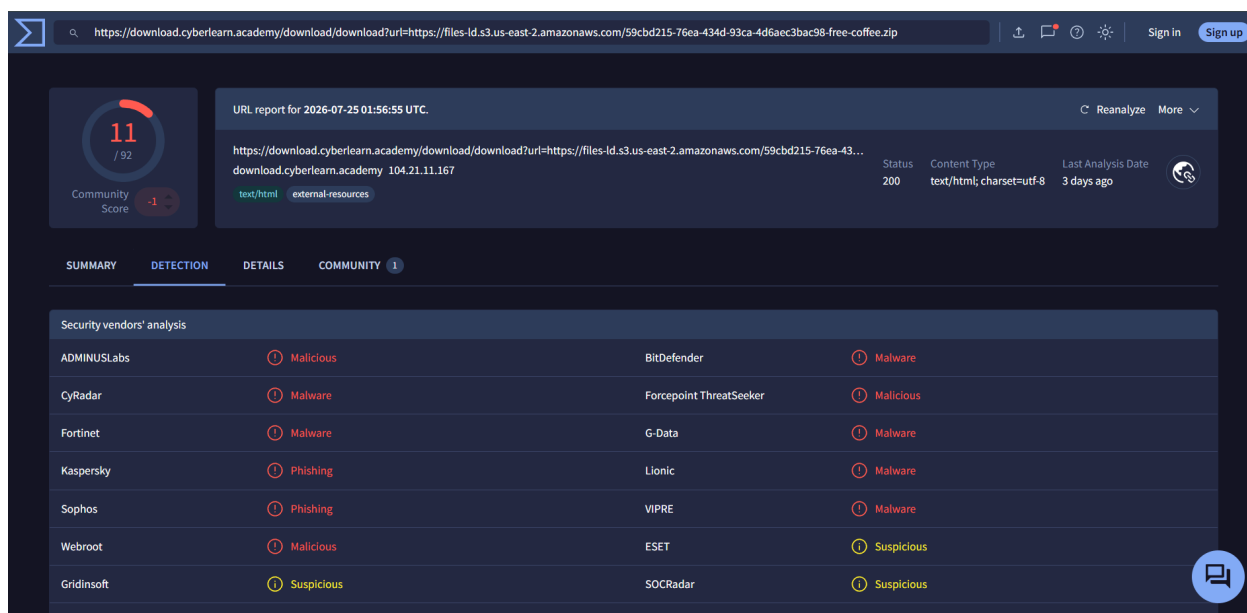
Using the **Email Security** section, I searched for the email associated with the alert and reviewed important indicators such as:

- Sender information
- Recipient
- Subject line
- Email headers
- URLs
- Attachments

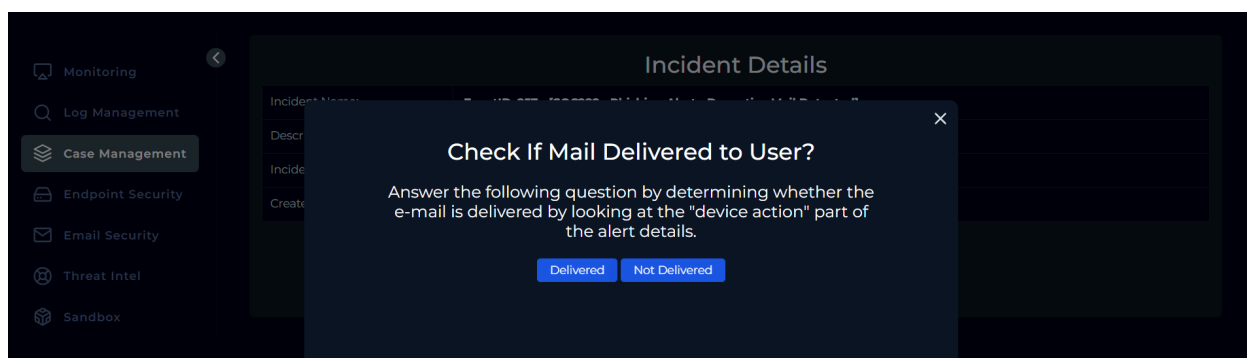


The email was found to contain an attachment named **free-coffee.zip** with the subject **"Free Coffee Voucher"**, both of which aligned with common phishing techniques that use enticing offers to trick users into opening malicious files.





Since the **URL was identified as malicious and associated with malware**, the next step was to verify whether the email had been successfully delivered to the recipient. This helps determine if the user was potentially exposed to the threat.



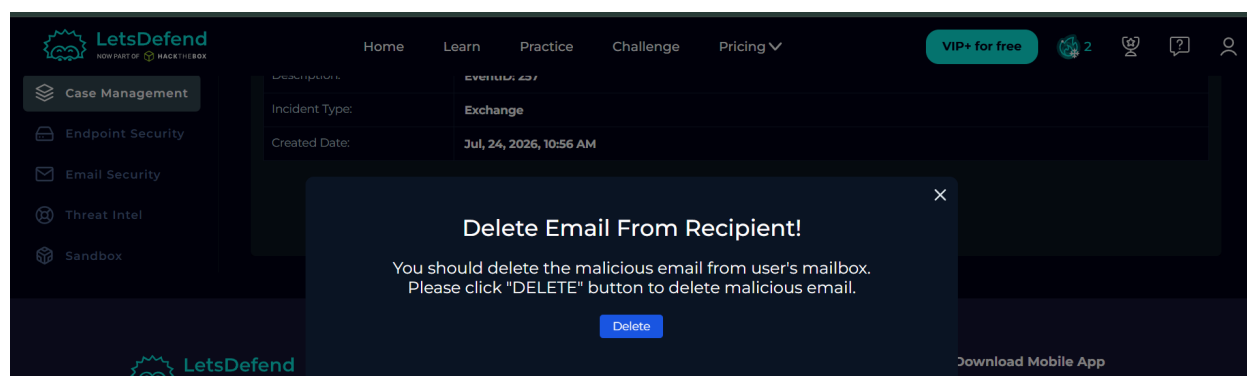
The investigation also confirmed the email's delivery status by reviewing the **Device Action** field in the alert details. Since the action was marked as **Allowed**, it indicated that the email had successfully reached the recipient's mailbox and could potentially be accessed by the user.

Email Containment

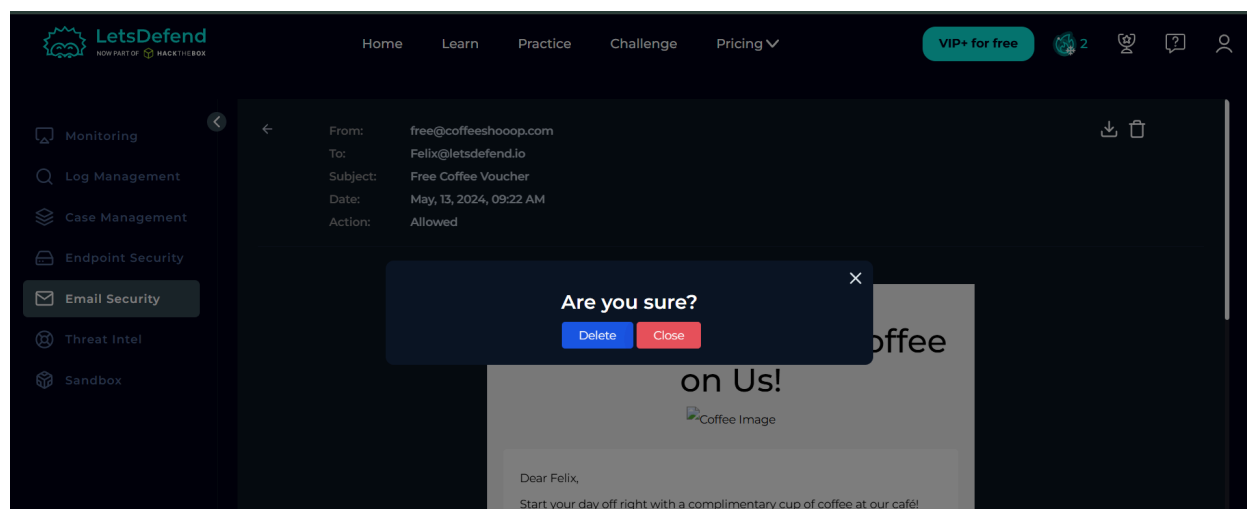
After confirming that the email was malicious and had been successfully delivered to the recipient, the next step was to prevent any potential user interaction.

Using the **Email Security** module, I searched for the suspicious email and reviewed its details to ensure it matched the alert. Once verified, I removed the email from the recipient's mailbox.

Deleting the email is an important containment measure because it reduces the risk of the user opening the attachment or clicking the malicious URL. Taking this action helps prevent the phishing attempt from leading to malware execution or further compromise.



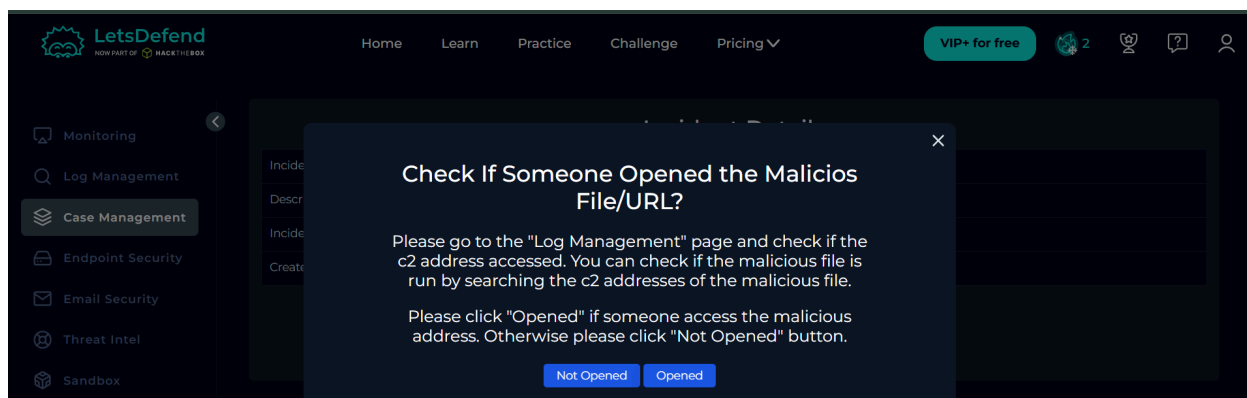
Navigate to the **Email Security** module and search for the suspicious email. After verifying that it matches the alert, open the email and click the **Delete** option in the top-right corner to remove it from the recipient's mailbox.



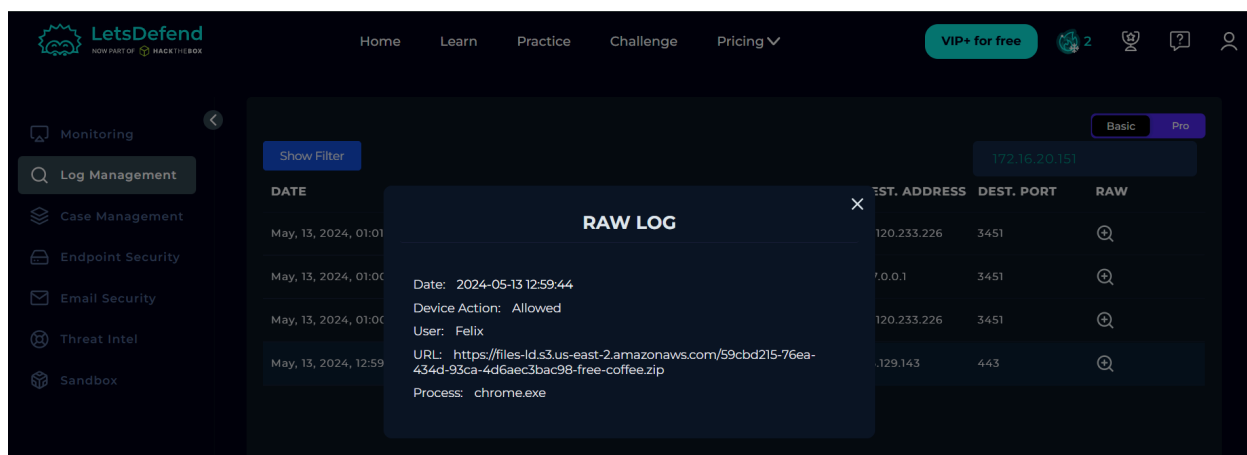
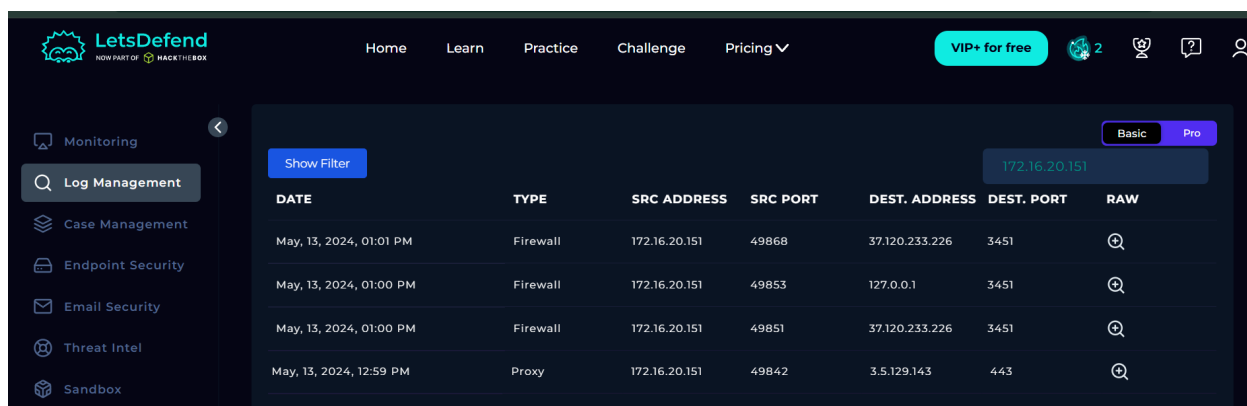
The malicious email was successfully removed from the recipient's mailbox, reducing the risk of user interaction and helping contain the threat before any further damage could occur.

5. Network & Log Analysis

The objective of this phase was to determine whether the user interacted with the phishing email and whether the malicious attachment resulted in any network activity.



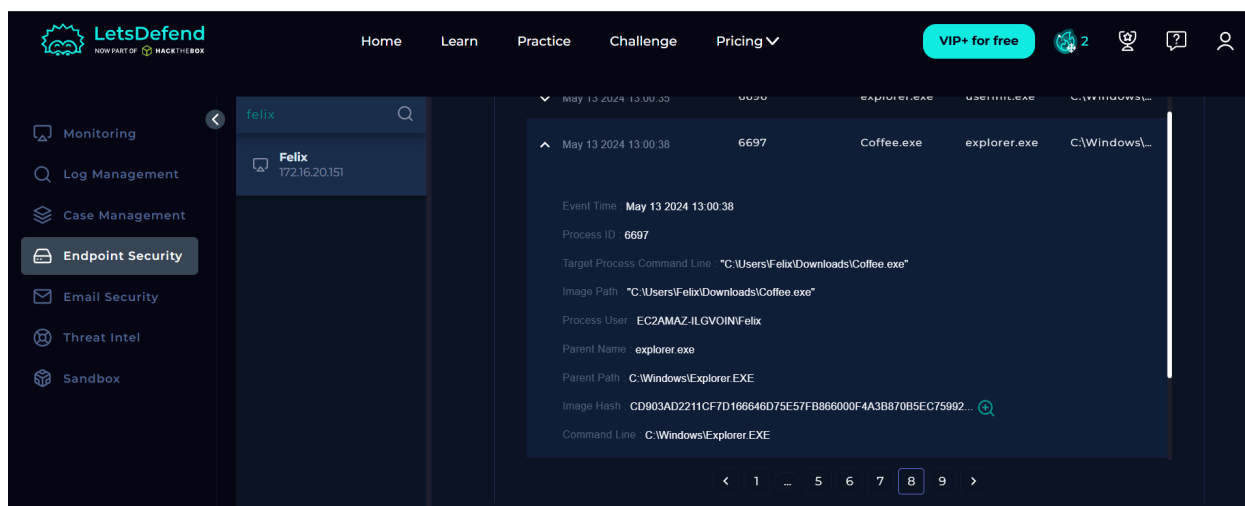
I first navigated to the **Endpoint Security** module to identify the affected endpoint and obtained its IP address (**172.16.20.151**). Using this IP, I searched the **Log Management** module to review the endpoint's network activity around the time the alert was generated.



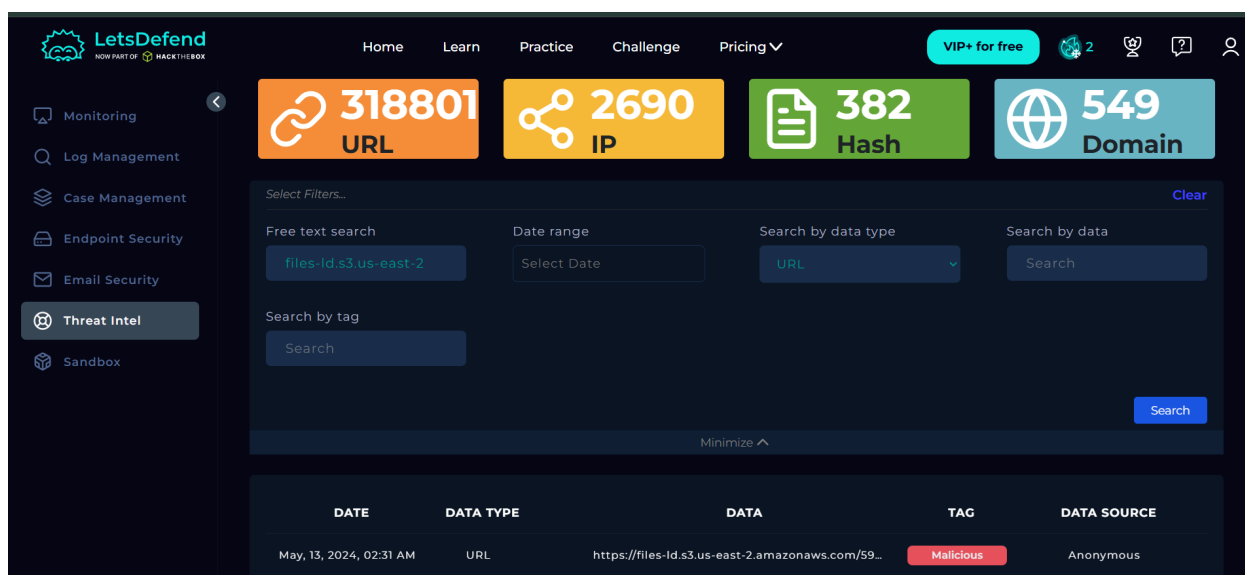
Using the **Endpoint Security (EDR)** module, I searched for the affected host and reviewed its host information, including the **hostname**, **IP address**, **operating system**, and **recent login activity**.

Based on the findings, the affected endpoint was **contained** to prevent further communication with the **Command-and-Control (C2)** server, minimize the risk of lateral movement, and stop any additional malicious activity.

During the investigation, I analyzed the source and destination IP addresses, timestamps, and raw log entries to identify any suspicious outbound connections. The logs confirmed that the user downloaded the malicious attachment on **May 13, 2024, at 12:59 PM**. Further analysis showed that the malware process **coffee.exe** established communication with the command-and-control (C2) server **37.120.233.226** over **port 3451**, indicating successful malware execution.



To validate these findings, I searched the identified indicators of compromise (IOCs) in the **Threat Intel** module. The lookup confirmed that the destination IP was associated with known malicious activity, reinforcing the evidence collected during the investigation.



Key Findings

- Affected Host IP: 172.16.20.151
- Malicious File Downloaded: Yes
- Download Time: May 13, 2024, 12:59 PM
- Malicious Process: coffee.exe
- C2 Server: 37.120.233.226
- Communication Port: 3451

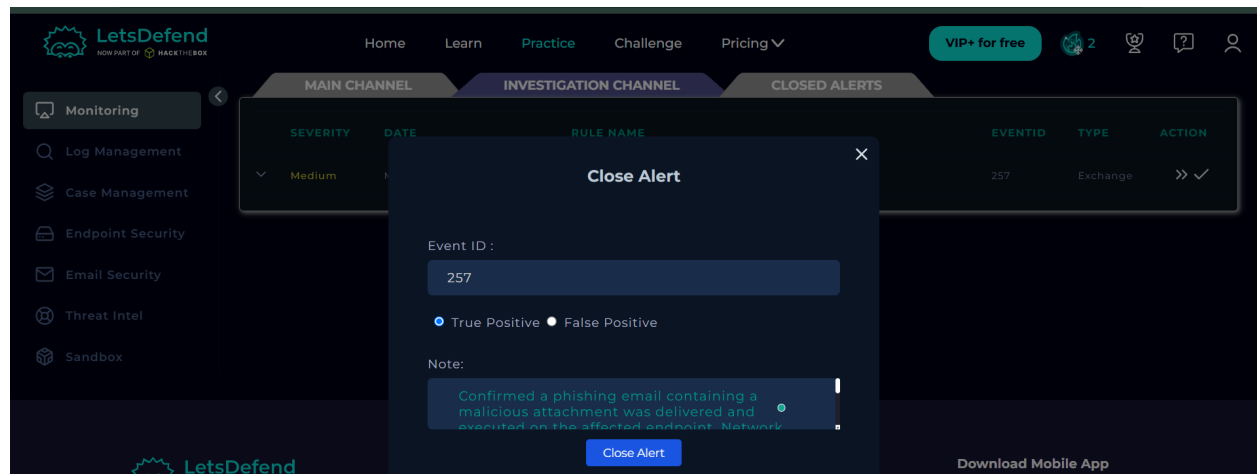
7. Investigation Outcome

Investigated **Event ID 257 – SOC282: Phishing Alert - Deceptive Mail Detected.**

The email contained a malicious attachment (**free-coffee.zip**) and was successfully delivered to the recipient (Device Action: Allowed). Log analysis confirmed the attachment was downloaded on **May 13, 2024, at 12:59 PM.**

The malicious process **coffee.exe** established communication with the C2 server **37.120.233.226** over **port 3451**. The phishing email was removed from the user's mailbox, and the endpoint was identified for containment.

Verdict: True Positive.



Skills Demonstrated

Throughout this project, I practiced several core SOC L1 responsibilities, including:

- SIEM Alert Investigation
 - Incident Triage
 - Phishing Email Analysis
 - Windows Log Analysis
 - Network Log Analysis
 - Endpoint Investigation
 - Evidence Collection
 - Playbook-Based Investigation
 - Threat Validation
 - Incident Documentation
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Tools Used

- LetsDefend SIEM
 - Email Security Module
 - Endpoint Security
 - Log Management
 - Case Management
 - Investigation Playbooks
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Key Takeaways

This project gave me practical experience in investigating phishing alerts using a structured SOC workflow. I learned how to analyze emails, correlate endpoint and log data, follow investigation playbooks, and determine whether an alert represents a real security incident.

Completing this investigation strengthened my understanding of the SOC incident response process and improved my ability to perform alert triage, evidence collection, and security analysis in a controlled environment.